

# EPAS18 Ultra ECU

## Data Sheet



The EPAS Ultra ECU is designed to work with the DCE EPAS01 Column Assist unit as well as OEM steering Columns from GM such as the Opel Corsa/Holden Barina MKB & C

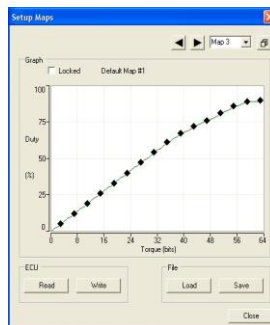


EPAS Ultra ECU

This Ultra ECU has been designed to handle transient currents of up to 80 amps. Connection is via 2 Autosport connectors.

- CAN 2.06 interface.
- 4 x Digital outputs.
- 2 x Speed inputs.
- 2 x Digital inputs.
- 3 x Analogue inputs.
- 5 User programmable steering maps.
- Internal heatsink temperature monitoring.
- External heatsink for maximum thermal performance.
- Internal temperature and peak current tell tales.
- Nominal Voltage 13.8v DC
- Weight: 1.05Kg

The CAN Bus can be used to monitor system performance or select the steering MAP. CAN control is also available for autonomous applications when using additional firmware.



Screen Shot of Steering Map Software

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### High Power Connector

| ECU Connector: AS016-08PN |              | Mating Connector: AS616-08SN       |        |
|---------------------------|--------------|------------------------------------|--------|
| Pin No.                   | Description  | Function                           | I/O    |
| A                         | Battery +12v | +12v from Battery via 80 amp Fuse* | Input  |
| B                         | Battery +12v | +12v from Battery Via 80 Amp Fuse* | Input  |
| C                         | Ground       | Battery Negative                   | Input  |
| D                         | Ground       | Battery Negative                   | Input  |
| E                         | Motor +12v   | Power To EPAS Motor                | Output |
| F                         | Motor +12v   | Power To EPAS Motor                | Output |
| G                         | Motor 0v     | Power To EPAS Motor                | Output |
| H                         | Motor 0v     | Power To EPAS Motor                | Output |

\* Pins A and B share the same fuse.

### Signal Connector

| ECU Connector: AS014-35PN |                                   | Mating Connector: AS614-35SN           |        |
|---------------------------|-----------------------------------|----------------------------------------|--------|
| Pin No.                   | Description                       | Function                               | I/O    |
| 1                         | +12v                              | Switched +12v rated at 1 amp           | Input  |
| 2                         | Ground                            | Battery Negative                       | Input  |
| 3                         | Steering Angle                    | 0-5v from Steering Angle Sensor        | Input  |
| 4-13                      | Not Used                          |                                        |        |
| 14                        | Sensor 0v                         | 0v for MAP Switch and Steering Sensor  | Output |
| 15                        | Sensor +5v                        | +5v for MAP Switch and Steering Sensor | Output |
| 16                        | Tx                                | RS232 Tx to Com Port                   | Output |
| 17                        | Rx                                | RS232 Rx from Com Port                 | Input  |
| 18                        | 0v                                | 0v to Com Port                         | Output |
| 19                        | CAN1 Hi (Internally linked to 29) | CAN Hi In                              |        |

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.... continued

| ECU Connector: AS014-35PN |                                   | Mating Connector: AS614-35SN  |        |
|---------------------------|-----------------------------------|-------------------------------|--------|
| Pin No.                   | Description                       | Function                      | I/O    |
| 20                        | CAN1 Lo (Internally linked to 30) | CAN Lo In                     |        |
| 21                        | Not Used                          |                               |        |
| 22                        | Not Used                          |                               |        |
| 23                        | Torque – A                        | Torque Sensor Signal A        | Input  |
| 24                        | Torque – B                        | Torque Sensor Signal B        | Input  |
| 25                        | Torque 0v                         | 0v for Torque Sensor          | Output |
| 26                        | Torque +5v                        | +5v for Torque Sensor         | Output |
| 27                        | Fault Lamp                        | Switched Ground to Fault Lamp | Output |
| 28                        | Switch Sig                        | Steering Map Selection Signal | Input  |
| 29                        | CAN1 Hi (Internally linked to 19) | CAN 1 Out                     |        |
| 30                        | CAN1 Lo (Internally linked to 20) | CAN1 Out                      |        |
| 31-37                     | Not Used                          |                               |        |

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### DIMENSIONS (MM)

